Reinforcement Learning for Hydrothermal Scheduling

Escaping the Convexity Trap of SDDP

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Course Project: Stochastic & Dynamic Programming

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The Domain: Hydrothermal Power Systems

- **Context:** The system operator must manage multiple hydroelectric reservoirs and thermal power plants to satisfy electricity demand.
- **Decision Dynamics:** Decisions are made sequentially at a monthly time scale over a multi-year horizon.
- **The State & Uncertainty:** In stage t, the operator observes the data realization ξ_t (e.g., water inflows) and takes an action based on the previous stage's decision x_{t-1} (reservoir volumes).
- **Objective:** Minimize expected long-term operational costs. Hydro generation is "free" but limited by water; thermal generation is reliable but highly expensive.

The Brazilian Hydrothermal System Architecture

- **Scale:** Uniquely hydro-dominated (∼75% of installed capacity), featuring complex cascades of large reservoirs with multi-year regulation.
- **Network Topology:** The grid is modeled as a graph with four primary generation nodes (Southeast, South, Northeast, North) and one transshipment node.
- **Operational Objective:** Minimize the expected cost of operation along a long-term planning horizon (e.g., 120 months).
- **Cost Components:** The total objective cost is the sum of expensive thermal generating costs plus a massive penalty term reflecting energy shortages.

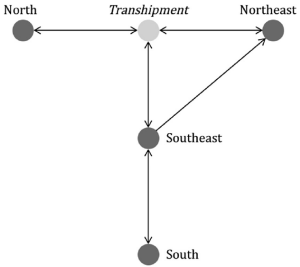

Figure: Interconnected power system

[1]

[1] A. Shapiro, W. Tekaya, J. P. da Costa, and M. P. Soares, "Risk neutral and risk averse Stochastic Dual Dynamic Programming method," *European Journal of Operational Research*, vol. 224, no. 2, pp. 375-391, 2013, doi: 10.1016/j.ejor.2012.08.022.

Mathematical Formulation (The MSSLP)

The long-term planning problem is formally cast as a Multistage Stochastic Linear Program (MSSLP):

$$\min_{A_1 x_1 = b_1} c_1^T x_1 + \mathbb{E}\left[\min_{\substack{B_2 x_1 + \\ A_2 x_2 = b_2}} c_2^T x_2 + \mathbb{E}\left[\cdots + \mathbb{E}\left[\min_{\substack{B_T x_{T-1} + \\ A_T x_T = b_T}} c_T^T x_T\right] \ldots\right]\right]$$

Defining the Variables and Parameters:

- x_t (Decisions): Hydro/thermal generation, reservoir volumes, spillage.
- c_t (Costs): Fuel costs for thermal units and deficit penalties.
- A_t, B_t (Constraints): Physical network topologies and transition dynamics.
- b_t (RHS): Stochastic elements (water inflows) and deterministic demand.

Unpacking the Physical Constraints

The matrices A_t and B_t enforce the physical reality of the grid:

1. **The Hydrological Mass Balance:**
   - Governs the transition of water states between months:

$$V_t = V_{t-1} + \text{Inflow}_t - \text{Turbined}_t - \text{Spilled}_t$$

   - Volume at stage t relies on the decision made at $t-1$ ($B_t x_{t-1}$) and the actions taken at stage t ($A_t x_t$).

2. **The Energy Balance Constraint:**
   - Must be strictly satisfied for each sub-system:

$$\sum \text{Hydro} + \sum \text{Thermal} + \text{Net Interchanges} + \text{Deficit} = \text{Demand}$$

Modeling Hydrological Uncertainty

- **The Stochastic Process (ξ_t):** The primary source of uncertainty is future water inflows, driven by unpredictable weather patterns. $\xi_t = (c_t, A_t, B_t, b_t)$, where b_t contains the random inflows.
- **Time Series Modeling:** Inflows exhibit strong seasonality and auto-correlation (a wet month usually follows a wet month). They are typically modeled as a Periodic Auto-Regressive ($PAR(p)$) process.

The SDDP Algorithm: Mechanics

Stochastic Dual Dynamic Programming (SDDP) avoids solving the full exponential scenario tree by building a piecewise linear approximation of the expected future cost function ($\mathcal{Q}_{t+1}$).

Iterative Process:
Sample M random inflow scenarios from the distribution.

- **Forward Pass (Trial Decisions):**
  - Traverse time $t = 1 \dots T$, solving the dispatch problem using current approximations to generate trial state decisions $\bar{x}_t$.
- **Backward Pass (Adding Cuts):**
  - Traverse backward $t = T \dots 2$. At each trial state $\bar{x}_{t-1}$, solve the problem for multiple inflow realizations.
  - Extract the dual variables of the mass balance constraints.
  - Construct supporting hyperplanes (e.g. Benders cuts) of the form $\theta_t \geq \alpha_t + \beta_t^\top x_{t-1}$ to iteratively refine the lower bound of the cost-to-go function.

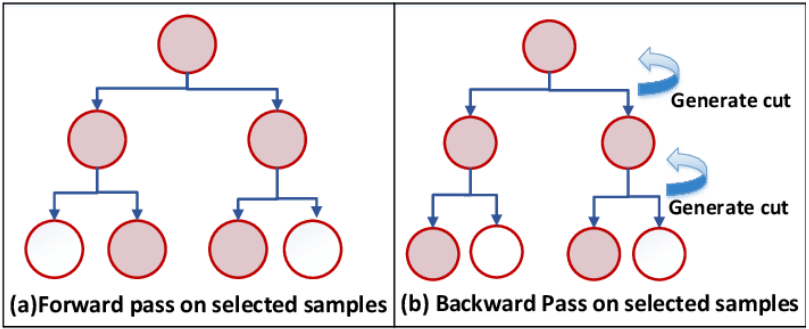

Figure: SDDP Algorithm[2]

[2] S. Zhang, Y. Cheng, and L. Cheng, "Coordinated Control of PEV and PV-based Storage System under Generation and Load Uncertainties," in *2018 IEEE Industrial Applications Society Annual Meeting (IAS)*, Portland, OR, USA, 2018, pp. 1-8, doi: 10.1109/IAS.2018.8544636.

Limitations of SDDP (The Convexity Trap)

While SDDP guarantees convergence to the optimal policy for linear problems, it imposes strict mathematical limitations on physical modeling:

- **Curse of Dimensionality via Augmentation:**
  - To use SDDP with auto-correlated $PAR(p)$ inflows, the operator must artificially augment the state vector x_t to include past inflows $(b_{t-1}, b_{t-2}, \dots)$.
  - SDDP fundamentally requires *stagewise independence* (the randomness at t cannot depend on $t-1$).
- **The Convexity Requirement:** SDDP requires the objective function and feasible region to be strictly convex.
- **Ignoring Physical Reality:** To maintain convexity, modelers must:
  - **Linearize Hydro Generation:** Entirely ignore the real turbine physical rule (where turbine efficiency non-linearly depends on the water height/pressure in the reservoir).
  - **Relax Unit Commitment:** Ignore or heavily approximate the binary, non-convex start-up and shut-down costs of thermal power plants.

Reinforcement Learning (RL) as the Solution

- **Why RL?** RL formulates the same sequential decision-making process as a Markov Decision Process (MDP) but does not require building piecewise linear Benders cuts of the cost-to-go function.
- **Handling Non-Convexity:** RL utilizes neural networks to approximate the policy directly from interactions with a non-linear simulation.
- **The Physical Advantage:** We can model the hydro physical rule and non-convex thermal constraints in the environment's transition function, allowing the RL agent to learn a policy that follows the true physics of the system.

RL Setup

State: Reservoir volumes

Action: The amount of hydro generation, thermal generation, and spillage

Reward: Minimize the cost

RL vs. SDDP Comparison Summary

Feature	SDDP (Baseline)	RL (Proposed)
Model Requirement	Strict linear/convex models	Black-box, non-linear simulators
Solution Format	Piecewise linear cuts ($\theta_{t+1} \geq \alpha + \beta^\top x_t$)	Neural Network Policy
Statistical Bounds	Yes (Upper/Lower bounds)	No (Convergence to local optima)

Proposed Methodology for the Project

- **Algorithm:** Implementing a continuous control RL algorithm (e.g., Actor-Critic) capable of handling high-dimensional action spaces.
- **Evaluation:** Comparing the RL method's results against a SDDP baseline model.
- **Expected Outcome:** RL should demonstrate superior policy performance in realistic scenarios where non-convexities heavily penalize the SDDP's linear approximations.